



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SMART ATTENDANCE SYSTEM USING FACE RECOGNITION

ABSTRACT:

The Smart Attendance System using Face Recognition is an automated and intelligent solution designed to overcome the limitations of traditional attendance methods, which are often time-consuming, error-prone, and vulnerable to manipulation. This system leverages advanced face detection and facial recognition techniques to accurately identify students and mark attendance in real time, even in large classroom environments. By eliminating the need for manual roll calls or physical interaction, the proposed system enhances efficiency, reliability, and transparency in attendance management within educational institutions.

During the enrolment phase, high-quality frontal facial images of students are captured under controlled conditions. These images are processed to extract unique facial features, which are transformed into numerical representations known as facial embeddings. Each embedding uniquely characterizes an individual's facial structure and is securely stored in a database along with corresponding student information such as roll number and batch details. This structured storage enables fast and reliable retrieval during the recognition process while ensuring data consistency and integrity.

In the attendance phase, a single group photograph of the classroom is captured using a standard camera. The system employs the RetinaFace algorithm to detect all faces present in the image, effectively handling variations in lighting, facial orientation, distance, and partial occlusion. Once detected, individual faces are cropped and passed to a custom-built face recognition module. This module generates embeddings for each detected face and compares them with the stored embeddings using similarity measures such as cosine similarity or Euclidean distance. Based on this comparison, the system accurately identifies students and automatically records their attendance with precise date and time stamps.

Overall, the Smart Attendance System provides a secure, scalable, and privacy-conscious solution that significantly reduces manual effort and administrative overhead. By offering accurate, fast, and automated attendance marking, it demonstrates the practical application of artificial intelligence and computer vision in education and serves as a reliable technological advancement for modern academic environments.

TEAM MEMBERS:

GUIDE NAME: P PRANITHA

22NG1A0507

22NG1A0525

22NG1A0520

22NG1A0557